1) $x+2y+z=0 \longrightarrow x=-2y-z$

$y=a,\ z=b \rightarrow x=-2a-b$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2a-b \\ a \\ b \end{pmatrix} = a\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + b\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

let $\{\vec{u_1}, \vec{u_2}\}$ be an orthogonal set.

we could find values of $\vec{u_1}, \vec{u_2}$ as

$$\vec{u_1} = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

$$\vec{u_2} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} - \frac{\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}}{\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}} \cdot \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} - \frac{2}{5}\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{5} \\ \frac{-2}{5} \\ 1 \end{pmatrix}$$

then, we will divide u_1, u_2 by their lengths to make this set orthonormal.

$$\frac{\vec{u_1}}{\|u_1\|} = \frac{\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}}{\sqrt{4+1+0}} = \frac{\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}}{\sqrt{5}} = \begin{pmatrix} \frac{-2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ 0 \end{pmatrix}$$

$$\frac{\vec{u_2}}{\|u_2\|} = \frac{\begin{pmatrix} -1/5 \\ -2/5 \\ 1 \end{pmatrix}}{\sqrt{\frac{1}{25}+\frac{4}{25}+1}} = \frac{\begin{pmatrix} -1/5 \\ -2/5 \\ 1 \end{pmatrix}}{\frac{\sqrt{30}}{5}} = \begin{pmatrix} -\sqrt{30}/25 \\ -2\sqrt{30}/25 \\ \sqrt{30}/5 \end{pmatrix}$$

so, the orthonormal basis is

$$\left\{ \begin{pmatrix} \frac{-2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{-\sqrt{30}}{25} \\ \frac{-2\sqrt{30}}{25} \\ \frac{\sqrt{30}}{5} \end{pmatrix} \right\}$$

3)

a) $H = I - 2uu^T \rightarrow H^2 = (I - 2uu^T)^2$

$H^2 = I^2 - 4\underbrace{uu^T}_{1} + 4\underbrace{uu^T}_{1}\underbrace{uu^T}_{1} = H^2 = I^2$

$I^2 = I$ so $H^2 = I$

b) $H^T = (I - 2uu^T)^T \rightarrow H^T = I^T - 2u(u^T)^T \rightarrow H^T = I^T - 2u^Tu$

$I^T = I$ so $H^T = I - 2uu^T$

then $H^T = H$, also $H^T \cdot H = I$

that means H is an orthonormal matrix.

c) $H \cdot u = (I - 2uu^T) \cdot u$

which equals to $u - 2uu^Tu = u - 2u = -u$

so the eigenvalue is $\lambda = -1$

4)

a) $[A \vdots b]$ augmented matrix involves the whole pivot column in matrix A. So the system is consistent, which means it has solutions.

A is a $n \times n$ matrix, its rank is r.

So the dimension equals to $\boxed{n-r}$

So, we could say $Ax=b$ linear equation set can be solved in this case.

b) The solution space of equation $x+3y-5z+t=5$ does not form a vector space because zero vector $(0,0,0,0)$ is not an element of solution space.

$x+3y-5z+t=5$ let's leave x alone at left

$x = 5-3y+5z-t$

so $\begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} 5-3y+5z-t \\ y \\ z \\ t \end{pmatrix}$ solution space has 3 dimensions

6) $\det\left(\begin{bmatrix} a_1 & a_1 & a_1 & \cdots & a_1 \\ 1 & a_2 & 1 & \cdots & 1 \\ 1 & 1 & a_3 & \ddots & 1 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & a_n \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 1 & 1 & \cdots & 1 \\ 1 & 1 & 1 & \ddots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & 1 \end{bmatrix}\right) =$

$$\det\left(\begin{bmatrix} a_1+1 & a_1+1 & a_1+1 & & a_1+1 \\ 2 & a_2+1 & 2 & \cdots & 2 \\ 2 & 2 & a_3+1 & \ddots & 2 \\ & \vdots & & \ddots & \vdots \\ 2 & 2 & 2 & \cdots & a_n+1 \end{bmatrix}\right) =$$

$$\begin{vmatrix} a_1+1 & a_1+1 & a_1+1 & & a_1+1 \\ 2 & a_2+1 & 2 & \cdots & 2 \\ 2 & 2 & a_3+1 & & 2 \\ & \vdots & & \ddots & \vdots \\ 2 & 2 & 2 & \cdots & a_n+1 \end{vmatrix}$$

so, we could find determinant by multiplying the diagonal values.

$$\det A = (a_1+1).(a_2+1).(a_3+1)\ldots(a_n+1)$$